



1) Consent-First Banking Authorization Gateway (ZK “greenlight” before settlement)

Concept: a drop-in authorization service where **every payment is pre-authorized by a SpendProof + ConsentReceipt**, producing an on-chain ProofVerified “authorization receipt” that banks/PSPs can settle against—**without ever receiving PII**.

Why it’s novel: it decouples *cryptographic authorization* from *financial settlement* while keeping compliance evidence intact.

What’s already in the docs: SpendProof public signals + ConsentReceipt + ProofVerified events, plus audit outputs like ProofReceipt/AuditTrailRoot.

MVP: browser proof generation + verifier contract + an API that returns a “settle-now receipt” object.

2) ZK-KYC Onboarding as a Service (admit customers via commitment, not documents)

Concept: a regulated onboarding workflow where users generate a **KYCProof** and the institution issues a detached **Attestation**, with CustomerRoot registered on-chain.

Why it’s novel: compliance can be proven while minimizing sensitive data retention risk.

What’s in the docs: KYCProof signals, attestation formats, CustomerAdmitted event, “circuits present / flow needs live execution” notes.

MVP: “KYC issuer” service + wallet UI that produces KYCProof and stores only commitments.

3) Zero-Disclosure Sanctions/AML Screening Layer (prove non-membership)

Concept: a screening module where the authority publishes a **SanctionsRoot**, and users prove **non-membership** (or membership) via ZK, optionally backed by signed ScreeningAttestation.

Why it's novel: it's "prove a negative" compliance with near-zero disclosure.

What's in the docs: ScreeningProof / SanctionsRoot / ScreeningAttestation patterns.

MVP: sanctions root publisher + client proof + bank verifier endpoint.

4) Healthcare Consent Mesh + Composable Modules (HIPAA-grade auditability, no PHI on-chain)

Concept: treat **consent commitments as the composable "root capability"**, then let data access, attestations, e-prescriptions, and device telemetry all link to it.

Why it's novel: consistent "lawful consent" enforcement across heterogeneous healthcare actions, without leaking identity or PHI.

What's in the docs: module rules ("no PHI on-chain", client-side first, nullifiers, drift bounds) + consent/data access/e-Rx details.

MVP: ConsentProof + FHIR pointer access relay (no logging PHI) + on-chain audit events.

5) "Silent-by-Default" Proof Runner (CSL gatekeeper that can refuse to publish)

Concept: a runtime that **evaluates CSL policies locally** and only submits transactions when prime gate + drift + policy commutation succeed; otherwise it takes the **silent path** and emits no on-chain trace.

Why it's novel: a practical "non-surveillance" enforcement mode: *compliance can block publication*, not just punish after the fact.

What's in the docs: prime/drift gates + CSL silent path; "Deploying with Silence" and PLIC packaging.

MVP: a local policy oracle + proof wrapper that outputs either `{submit}` or `{silent}` with a signed local audit bundle.

6) DIN-Ready Policy-Aware RPC Middleware (“Infra Evidence Adapter” + Archivum receipts)

Concept: a middleware layer where **ProofRequest** includes **transport intent** and **PolicyVerdict** returns **infra observations**; optionally ingest DIN AVS metrics later without changing higher-level logic.

Why it’s novel: makes RPC reliability *auditable and policy-visible* while keeping “lawfulness not delegated to infra.”

What’s in the docs: RpcProfile abstraction, ProofRequest/PolicyVerdict shapes, infra evidence adapter idea, and ArchivumReceipt extension.

MVP: “DIN-switchable” client/provider factory + logging of transport evidence into Archivum artifacts.

7) Λ-Trace IP Passport + Licensing Marketplace (proof-anchored invention + evaluation)

Concept: package a project/protocol/IP asset into an ERC-721 “passport” that anchors **workHash**, **ltraceHash**, **archivumCidHash**, then issue **license tokens** and **contribution badges** (ERC-1155), and time-bound **lab passes** (ERC-4907).

Why it’s novel: turns R&D/IP governance into a cryptographically-linked trail of evaluation, evidence, and permissions—without reducing contributors to pure speculation.

What’s in the docs: token field schemas + contribution/license/lab-pass structures + canonical hashing.

MVP: “proposal evaluation → Λ-Trace hash → mint passport → issue lab pass” flow.

8) Prime-Epoch Key Rotation & Prime-Triggered Encryption SDK (PLEM)

Concept: a usable crypto SDK where key rotation and encryption triggers occur at **prime epochs** (e.g., prime block heights), integrated into app storage and PLIC notarization workflows.

Why it’s novel: turns timing into a *verifiable, non-predictable but checkable* security schedule.

What’s in the docs: PLEM key rotation at prime epochs + prime-tied triggers; PLIC/IPFS notarization patterns.

MVP: a library that exposes `encryptAtPrimeEpoch()` + an on-chain anchor of the epoch seal.

9) Prime-Gated QIoT Device Activation & Firmware Updates (enterprise device security)

Concept: manage IoT/QIoT devices so that activation, firmware updates, and secure emission can only happen when **IsPrime(epochPrime)** (plus optional drift bounds).

Why it's novel: sparse, number-theoretic scheduling reduces the continuous attack surface to discrete auditable windows.

What's in the docs: QIoT prime-epoch emissions and the prime-gated device management rationale.

MVP: a device agent + management console that issues “prime window” activation certificates and logs proofs/receipts.

10) Verifiable Knowledge Archive + Ξ Semantic Graph (anti-drift documentation + provenance)

Concept: build an “Archivum for knowledge” where each artifact is assigned a **prime identity / prime factors**, stored in an append-only provenance ledger (Merkle-CRDT), then connected via a constrained semantic graph (derives_from/proves/contradicts/etc.) with invariants (entropy bounds, drift bounds, provenance completeness).

Why it's novel: a knowledge base that's *structurally verifiable* and can quarantine “tainted” content automatically.

What's in the docs: Λ^p registry + Merkle-CRDT ledger + Ξ semantic graph + compiler pipeline/invariants; and prime-identity archives preventing semantic drift.

MVP: ingest $\rightarrow$ canonicalize $\rightarrow$ hash $\rightarrow$ emit edges $\rightarrow$ store Merkle receipts; add a “quarantine” flag when invariants fail.

11) CSL-Governed AI Decision Gate (pre-execution bias/stability interception)

Concept: apply CSL as a runtime gate for high-stakes AI (loan approvals, triage, underwriting): monitor “prime-indexed tensor” signatures representing sensitive attributes and block/flag updates that violate ethical invariants.

Why it's novel: moves from post-hoc audits to *intrinsic enforceability* at the computation boundary.

What's in the docs: CSL integrated into AI runtime to prevent operations that would violate

ethical invariants.

MVP: a policy wrapper that requires a “CSL pass” attestation before deploying a model update (even before full ZK proofing).

12) “Lawful Self-Update / Self-Replication” Guardrails (for agents and protocol upgrades)

Concept: a safe self-upgrade framework where replication/self-modification is only allowed if it preserves prime-indexed audits, provides formal witnesses (cohomology/coding constraints), and supports rollback on violation.

Why it’s novel: a mathematically-framed upgrade safety regime that’s compatible with auditability.

What’s in the docs: cohomology-backed admissibility, error budgets, rollback workflow; plus runtime firewall/verifier concepts.

MVP: “upgrade proposal object → required evidence bundle → allow/deny → rollback trigger” (even if witnesses start as structured attestations before full formalization).